

Md Shaqib Hussain

Software Development Engineer

Bengaluru, India | ☎ +91-6203786026 | ✉ mdshaqibhussain5@gmail.com

🌐 linkedin.com/in/md-shaqib-hussain | 🐙 github.com/Shaqib27

PROFESSIONAL SUMMARY

Computer Science Engineer with a Post Graduate Diploma in Big Data Analytics (AI & ML) from C-DAC and hands-on experience in software development, C/C++, Python, data structures, algorithms, Linux, and machine learning. Strong foundation in object-oriented programming, problem solving, algorithmic thinking, and scalable application development, with experience building and deploying data-driven software solutions. Familiar with debugging, Git, AWS, and distributed processing using Apache Spark. GATE 2025 qualified with strong analytical and problem-solving foundations, seeking to contribute to software development and optimization in a product-focused R&D environment.

TECHNICAL SKILLS

Programming Languages: C, C++, Python, SQL

Core Computer Science: Data Structures & Algorithms, Object-Oriented Programming, Problem Solving, Algorithm Optimization, Complexity Analysis, Debugging

Operating Systems & Platforms: Linux, Shell Scripting, AWS EC2

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Supervised Learning, Unsupervised Learning, Classification, Regression, Feature Engineering, Model Evaluation

Deep Learning: CNN, RNN, LSTM, Transfer Learning, Transformers, Attention Mechanisms

Big Data & Distributed Computing: Apache Spark, PySpark, Hadoop

Databases: MySQL, SQL Joins, Aggregations, Window Functions, Data Warehousing

Cloud & Deployment: AWS EC2, AWS S3, FastAPI, Docker

Development Tools: Git, GitHub, VS Code, JupyterLab

EXPERIENCE

Graduate Engineer Trainee – Data Analyst

Mar 2026 – Aug 2026

SINCLUS Engineering & Consulting Pvt. Ltd. (BPCL Project)

Mumbai, India

- Queried and consolidated operational data from multiple source systems using SQL and Python, supporting reliable data processing and analysis workflows.
- Performed statistical analysis and exploratory data analysis (EDA) on production datasets to identify trends, anomalies, and data quality issues.
- Built and maintained Power BI dashboards tracking 10+ key operational metrics, improving visibility for cross-functional stakeholders across 3 departments.
- Automated recurring data extraction and cleaning workflows using Python and Excel, reducing manual processing time and reporting errors.
- Collaborated with cross-functional teams to understand business requirements and translate them into analytical solutions and deliverables.

PROJECTS

IntelliScan: AI-Assisted Cancer Screening System | *TensorFlow, Keras, CNN, Transfer Learning, Scikit-learn* **Feb 2026**

- Designed and trained a CNN-based deep learning model using TensorFlow and Keras for automated binary cancer detection and multi-class classification across 5+ cancer categories.
- Applied transfer learning (VGG-style architecture) and systematic hyperparameter tuning to improve model generalization on imbalanced classes, achieving 91% validation accuracy and 0.93 AUC-ROC.
- Implemented data augmentation, dropout regularization, and early stopping to prevent overfitting; evaluated outputs using precision, recall, F1-score, and confusion matrices – lifting classification accuracy by 18%.
- Leveraged Apache Spark (PySpark) for distributed preprocessing and feature extraction on 50,000+ images, enabling reproducible and scalable model training workflows.

Air Quality Forecasting System | *Python, Scikit-learn, LightGBM, XGBoost, Statsmodels, Streamlit, AWS EC2* **Jan 2026**

- Built an end-to-end ML pipeline for multi-hour AQI forecasting – covering data ingestion, exploratory data analysis (EDA), time-series feature engineering, model training, evaluation, and AWS deployment.
- Engineered lag variables, rolling statistics, and interaction features using Pandas and Statsmodels to capture complex temporal dependencies across 5+ air pollutant types.
- Trained and cross-validated LightGBM and XGBoost ensemble models across 10+ forecast horizons, achieving R^2 of 0.89 and RMSE of 12.4 – a 22% improvement over baseline.
- Deployed an interactive Streamlit application on AWS EC2 enabling real-time AQI forecasts from user-uploaded CSV data, validated by 20+ test users during evaluation.

EDUCATION

Post Graduate Diploma in Big Data Analytics (AI & ML) <i>CDAC – Centre for Development of Advanced Computing, Pune</i>	Percentage: 86.88% <i>Aug 2025 – Feb 2026</i>
--	---

Bachelor of Technology in Computer Science and Engineering <i>Heritage Institute of Technology, Kolkata</i>	CGPA: 7.54 <i>Aug 2021 – Jun 2025</i>
---	---

CERTIFICATIONS & ACADEMIC HIGHLIGHTS

GATE 2025 – Computer Science and Engineering (Qualified)	Mar 2025
---	-----------------

CDAC PG Diploma in Big Data Analytics (AI & ML) – 86.88%	Feb 2026
---	-----------------

EXTRACURRICULAR ACTIVITIES

National Service Scheme (NSS) <i>Coordinator – Heritage Institute of Technology, Kolkata</i>	Aug 2021 – Jun 2025
--	----------------------------

- Led campus and community awareness drives; coordinated educational support programs in local schools in Kolkata, benefiting 100+ students over 4 years.